

Aravindan Rameshbabu

SENIOR DATA & AI ENGINEER

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Senior Data & AI Engineer with 5+ years spanning large-scale cloud data platforms and applied GenAI systems. Equally at home architecting an open-source data lake on EKS (Spark, Iceberg, Trino, Airflow) and building RAG pipelines with ChromaDB, MCP-based tool orchestration, and open-source LLMs. Led a 5-member team end-to-end and consistently recognized for technical ownership and delivery.

TECHNICAL EXPERTISE

AI / GenAI: RAG Systems, MCP (Model Context Protocol), LLMs, ChromaDB, SeqGAN, Ollama

Data & Streaming: Apache Spark (PySpark, Spark-SQL, Spark Streaming), Apache Kafka, Hadoop, Airflow, Trino, Dremio, Iceberg, Nessie

Cloud & Infra: AWS (EC2, S3, Glue, IAM), Kubernetes, Databricks, GitHub

Databases: MongoDB, MySQL, PostgreSQL, Cassandra

Languages & Patterns: Python, SQL, Scala, Multi-tenant ETL Orchestration, Medallion Architecture

WORK EXPERIENCE

Senior Data Engineer (SDE 2) — M2P Fintech

Feb 2025 – Present

[Chennai, India](#)

- Architected a scalable open-source data lake on EKS integrating Spark, Iceberg, Trino, Airflow, Nessie Catalog, and blob storage, significantly reducing data access latency and storage costs.
- Designed and automated ETL workflows using Airflow DAGs with Spark and Iceberg for data processing, Trino for federated querying, and Nessie Catalog for metadata management.
- Extended the platform into an agentic tech stack, driving ETL pipelines via MCP (Model Context Protocol) to let LLM agents trigger and monitor data workflows.
- Built SCD Type 2 history pipelines on Apache Iceberg with a Hot/Cold table architecture and dynamic watermarking, enabling full fintech audit trails and KPI roll-ups over impacted date partitions.
- Automated serverless ETL with AWS Glue and Lambda, cutting manual schema management by 40% and enabling real-time data validation.
- Secured all workloads using IRSA (IAM Roles for Service Accounts) for Spark, Airflow, and Trino; integrated Teams webhook alerting for instant pipeline failure notifications.
- Led a 5-member team to deliver the end-to-end data lake platform on time, earning recognition for technical ownership and cross-functional leadership.

Data Engineer — Tata Consultancy Services

Apr 2021 – Feb 2025

[Chennai, India](#)

- Developed and optimised PySpark applications on a Kubernetes-managed Spark cluster for large-scale advanced analytics.
- Automated ETL workflows using Apache Airflow DAGs on Kubernetes, improving pipeline efficiency and resource utilisation.
- Migrated Oracle RDBMS to GCP Cloud using PySpark and Spark-SQL, reducing processing time and improving data reliability.
- Optimised storage and query performance by implementing partitioned Hive tables with compression across AWS S3 and GCP.

- Integrated Spark API and Kafka for real-time data processing; migrated VDS to Dremio for optimised federated data access.
- Recognised with 'Technical Excellence' and 'Inspiring Talent' awards for consistently delivering high-impact data engineering projects.

SELECTED PROJECTS

Prism — Agentic Control Plane for a Data Lakehouse · M2P Fintech

- Built a full-stack control plane hosting Airflow, Spark, Trino, and Unity Catalog to orchestrate ETL pipelines that ingest multi-source data into Iceberg and Delta tables.
- Made the entire pipeline MCP-friendly, letting users define source/destination connections and dictate table migrations through a single chat interface instead of manual config.
- Automated end-to-end orchestration — Airflow-driven DAGs trigger Spark jobs for ingestion, transformation, and load, from source connection to target table.

Synthetic Telecom Data & RAG/MCP Orchestration Platform · Tata Consultancy Services

- Built a SeqGAN architecture to generate high-fidelity synthetic telecom data, reducing model bias on imbalanced datasets.
- Developed a RAG system with ChromaDB vector embeddings and Ollama-powered open-source LLMs for fast, domain-specific intelligence retrieval.
- Exposed retrieval and data tools to LLMs via MCP (Model Context Protocol) servers, standardising how agents accessed telecom data sources.
- Orchestrated the full AI lifecycle — ingestion, embedding, deployment, and monitoring — via Apache Airflow.

Real-Time Location-Based Marketing · Tata Consultancy Services

- Used Spark Streaming to ingest real-time location data from IMS (ISP) towers and cross-validate against customer records to identify marketing targets.
- Stored recipient history in Cassandra NoSQL with a 6-month retention window; delivered filtered data to the Macy API in full GDPR compliance.

EDUCATION

B.E. Electronics & Communication Engineering — University College of Engineering Tindivanam, Anna University

2016 – 2020

- Graduated First Class with CGPA 7.05.
- Final-year project: Face Recognition Model using Python and OpenCV with Haar Cascade Classifiers — graded 'S' (Top Grade).
- Served as Chairman of the Student Cell; volunteered across multiple institutional events.

CERTIFICATIONS

- Azure Data Engineer Associate (DP-203)
- Data Streaming Engineer – Apache Flink, Confluent (Apr 2026 – Apr 2028)
- Linux Essentials (LE-01)
- Hands-on Approach to AI for Real-World Applications

AWARDS

- Technical Excellence — TCS, Aug 2022
- Inspiring Talent — TCS, Jun 2023